

PROJECT REPORT

VIT GUIDE – A CAMPUS INFORMATION SYSTEM

Student Name

Utkarsh Singh

Registration ID

26BAC10047

Programming Language

Python

Project Type

Console based Python Project

1. Introduction

VIT Guide is a simple Python console-based project created to display simple campus information. The application is menu-driven and has options to view selected campus places, hostel blocks, sports courts, shops, building information and a personal favourites list.

The project is of beginner level and demonstrates the basic Python programming concepts of lists, dictionaries, functions, loops, conditions, user input and basic input validation.

2. Objectives

To create a simple Python campus information guide.

To display information about selected campus locations and facilities.

Using lists and dictionaries to store data.

Using functions to break the program in to manageable sections.

Using loops and conditions to create a menu-driven interface.

To provide search/order features and manage a favourites lists.

3. Project Features

Special Block – Displays AB Dakshin, Bistro, Mayuri, Gym and Dentist – users can view menus, order simple items and add to favourites.

Boys Hostel – Displays Block 1 and Block 2 with their listed facilities.

Sports Courts – Lists Basketball, Volleyball, Football, Badminton and Tennis courts and adds to favourites.

Shops – Shows shop locations, item prices and provides a simple buying/order facility.

AB1 Building – Displays the teachers' cabin information and listed cafes.

Search – Searches stored campus information based on a word entered by the user.

My Favourites – Displays saved favourites and allows a user to remove a selected favourite.

4. Data Used in the Project

The program uses various dictionaries and lists to store the campus information. The source includes menus and prices for AB Dakshin, Bistro, Mayuri, Gym and Dentist and sports courts, shops, hostel blocks, shop items and AB1 information.

5. Python Concepts Used

Variables and assignment

Lists and list operations

Dictionaries and dictionary lookup

Functions and function calls

for loops

while loops

if/elif/else statements

User input using input()

Integer conversion using int()

String operations such as lower() and join()

Basic input validation using isdigit()

6. Important Functions

title(text) – prints a formatted heading.

ask_number(message) – keeps asking until the user enters a number.

pause() – waits for the user to press Enter.

show_list(items) – displays list elements with numbers.

add_favourite(name) – adds an item to favourites if it is not already present.

make_order(heading, menu) – displays a menu, accept quantities, calculates a total, prints a bill.

show_menu(place) – displays the stored menu and information for a selected place.

place_options(place) – provides options for a selected Special Block place.

special_block_section() – displays Special Block places.

hostel_section() – displays boys hostel blocks and facilities.

courts_section() – displays sports courts and handles favourites.

shops_section() – provides shop locations, item prices, buying options.

ab1_section() – displays AB1 building information.

search_section() – searches stored campus categories.

favourites_section() – displays and manages favourites.

main() – controls the welcome screen and main menu.

7. Program Flow

1. Start the program.
2. Ask the user for their name.
3. Display the main menu.
4. The user selects a numbered option.
5. The selected section opens.
6. The user can view information, order items, search, or manage favourites.
7. The program returns to the main menu until Exit is selected.
8. Display a thank-you message and end.

8. Main Menu

1. Special block
2. Boys hostel
3. Sports courts
4. Shops
5. AB1 building
6. Search
7. My favourites

0. Exit

9. Order and Billing System

The `make_order()` function is used for food ordering and shop-item buying. It displays available items and prices, accepts an item number and quantity, calculates the total, stores selected items in a bill list, and displays the final amount.

Basic calculation: $\text{Total} = \text{Item Price} \times \text{Quantity}$, added for each selected item.

10. Search System

The search section asks the user to type a name or word. It checks the entered text against Special Block places, sports courts, shops, boys hostel blocks, and shop items. The `lower()` method is used so the search is not dependent on uppercase or lowercase letters.

11. Input Validation

The `ask_number()` function checks numeric input using `isdigit()`. If the user enters a non-numeric value, the program displays an error message and asks again.

12. Testing

Test

Input/Action

Expected Result

Main menu

Enter a valid menu number

Corresponding section opens.

Invalid input

Enter letters where a number is required

Program asks for a number again.

Special Block

Select a place and show menu

Stored menu and information are displayed.

Ordering

Select item and quantity

Item is added and total is calculated.

Sports courts

Select a court

Court is displayed and can be added to favourites.

Search

Enter a stored word

Matching information is displayed.

Favourites

Add and remove an item

Favourites list is updated.

13. Limitations

The application is console-based and has no graphical user interface.

Campus information is stored directly in the Python source code.

There is no database or online connection.

Orders are simulated and are not sent to real shops or restaurants.

Search works only with information stored in the program.

14. Future Scope

Add a graphical user interface.

Store information in a database.

Add user login/accounts.

Add campus maps and navigation.

Add more buildings and facilities.

Add more detailed ordering and receipt features.

15. Conclusion

VIT Guide is a simple Python project that demonstrates how fundamental programming concepts can be combined to create a useful menu-driven application. It uses lists, dictionaries, functions, loops, conditions, user input, and basic validation. The project can be further developed into a larger campus information system.

16. Source File

This report is based on the submitted Python source file, utkarsh.py, which contains the VIT Guide data, menu sections, ordering functions, search functionality, favourites

management, and main program loop.